

Report on Industry Needs in Sustainable Textile and Fashion Materials

Executive Summary

The sustainable textile and fashion sector in Europe is undergoing transformative change driven by urgent environmental and social imperatives. Our comprehensive research—anchored by two surveys targeting universities and creative hubs/networks—reveals a rich, yet fragmented ecosystem of stakeholders whose needs, challenges, and aspirations intersect around knowledge accessibility, material innovation, and collaboration.

This report synthesizes the detailed stakeholder landscape—including academic institutions such as **Politecnico di Milano (IT)**, **Aalto University (FI)**, **Moholy-Nagy University (HU)**, creative hubs like **Lottozero Textile Hub (IT)**, **Textile Prototyping Lab (DE)**, and networks such as **European Textile Network (BE)** and **EURATEX (BE)**—and identifies critical gaps and opportunities. We align these insights with the **Imasus project's mission**: to facilitate sustainable innovation by creating a robust, accessible knowledge infrastructure and fostering cross-sector collaboration.

1. Introduction

Sustainability challenges in textile and fashion demand coordinated innovation across disciplines and industries. The Imasus project seeks to catalyze this by building an integrated digital platform that empowers all actors—from researchers to designers to manufacturers—to access verified material information, share expertise, and co-develop sustainable solutions.

To ensure Imasus addresses real-world needs, we conducted two complementary surveys:

- **Survey 1:** Engaged universities and educational institutions listed in the stakeholder file, such as **Aalto University (FI)**, **Politecnico di Milano (IT)**, **NABA (IT)**, and **Moholy-Nagy University of Art and Design Budapest (HU)**, focusing on academic research, pedagogy, and material science innovation.
- **Survey 2:** Engaged creative hubs, textile labs, institutional networks, and innovative enterprises such as **Lottozero Textile Hub (IT)**, **Textile Prototyping Lab (DE)**, **Waag Textile Lab (NL)**, and **Ecoalf (ES)**, working hands-on with textile and fashion sustainability.

This report merges insights from both, relating their unique and overlapping challenges, and positioning Imasus as a key enabler.

2. The Stakeholder Ecosystem

Our stakeholder landscape reveals a multi-layered network:

2.1 Academic Institutions (Universities and Schools)

- **Politecnico di Milano (IT):** Known for cutting-edge material research and design education, highlighting the need to better translate research outputs for practical industry application.
- **Aalto University (FI):** Emphasizes interdisciplinary approaches combining technology and sustainability, expressing interest in enhanced collaboration platforms to bridge academia and industry.
- **Moholy-Nagy University of Art and Design Budapest (HU):** Focuses on innovative textile materials and circular design, indicating a need for transparent material databases and accessible certification information.
- **NABA - Nuova Accademia di Belle Arti (IT):** Prioritizes embedding sustainability in design curricula, but faces challenges integrating up-to-date material science data into teaching.
- **Beuth Hochschule für Technik Berlin (DE):** Engaged in textile engineering research, seeking access to verified supplier information and sustainable material sourcing.

These institutions contribute deep scientific expertise but identify gaps in knowledge dissemination and practical industry engagement.

2.2 Creative and Textile Labs

- **Lottozero Textile Hub (IT):** Acts as a bridge between design experimentation and production but lacks scalable frameworks to extend innovations beyond pilot projects.
- **Textile Prototyping Lab (DE):** Focuses on prototyping sustainable materials but struggles with limited access to comprehensive, validated supplier and certification data.
- **Waag Textile Lab (NL):** Drives hands-on experimentation and educational outreach, requesting more open-access knowledge bases and modular learning materials.
- **Le Textile Lab (FR):** Highlights challenges in combining circular design principles with industrial scalability, underlining the need for better industry-academia linkages.

2.3 Networks and Industry Platforms

- **European Textile Network (BE):** Advocates for pan-European collaboration and improved knowledge standardization to reduce fragmentation across sectors.
- **EURATEX (BE):** Emphasizes the importance of transparent supply chains and certification harmonization to foster sustainable material adoption.
- **ACTE - European Association for Creative Textile and Fashion Enterprises (BE):** Supports creative SMEs but notes difficulties in accessing verified material databases and cost-effective circular solutions.
- **MAD Brussels (BE):** Focuses on facilitating sustainable innovation and cross-border partnerships within the creative industries.

2.4 Creative Hubs and Studios

- **Studio Sarmite (DE):** A small design studio that pioneers upcycling but faces knowledge gaps in sourcing sustainable certified materials affordably.
- **Ecoalf (ES):** A leader in circular fashion, emphasizing the need for transparency and traceability in supply chains to maintain consumer trust and scale innovations.
- **Labore (IT):** Highlights the importance of cross-disciplinary exchange between designers and technologists to foster innovative sustainable products.

3. Industry Needs and Challenges

Despite the diversity, both surveys highlighted converging needs:

3.1 Access to Reliable, Transparent Material Information

Universities like **Politecnico di Milano** and **Moholy-Nagy University** possess deep scientific knowledge but find it difficult to translate complex research into accessible, industry-applicable insights. Creative hubs such as **Studio Sarmite** and labs like **Lottozero Textile Hub** face barriers accessing verified, affordable material data due to fragmentation and inconsistent certifications.

Imasus Opportunity: Provide a unified, open-access platform aggregating peer-reviewed material data, verified supplier information, and certification details.

3.2 Material Sourcing and Validation

Sourcing genuinely sustainable textiles is hindered by inconsistent certifications and lack of centralized, trustworthy supplier databases, as reported by **Beuth Hochschule für Technik Berlin** and **Textile Prototyping Lab**.

Imasus Opportunity: Integrate certified supplier directories and standardized sustainability metrics into the platform.

3.3 Bridging Theory and Practice

Creative hubs, including **Waag Textile Lab** and **Le Textile Lab**, express difficulty in scaling lab innovations due to insufficient industry infrastructure and lack of practical know-how for circular economy implementation.

Imasus Opportunity: Facilitate knowledge transfer via training modules, real-world case studies, and collaborative innovation projects.

3.4 Cost and Market Accessibility

Smaller studios like **Studio Sarmite** and emerging enterprises noted that sustainable materials remain costly, limiting market access and scalability.

Imasus Opportunity: Highlight cost-effective alternatives and support pilot initiatives demonstrating scalable circular business models.

3.5 Collaborative Networking

Networks such as **European Textile Network** and **MAD Brussels** underscore the need for stronger cross-sector collaboration to accelerate sustainable innovation.

Imasus Opportunity: Build community tools and event facilitation features to foster ongoing collaboration and resource sharing.

4. The Imasus Project's Strategic Role

Imasus aims to become the **digital backbone** for sustainable textile innovation by:

- Creating an **open, dynamic database** aggregating validated scientific material data, sustainability certifications, and supplier information from academic partners like **Aalto University**, **Politecnico di Milano**, and creative labs such as **Lottozero** and **Waag Textile Lab**.
- Offering **educational resources** tailored to various expertise levels, co-developed with universities (**NABA**, **Moholy-Nagy University**) and industry networks (**EURATEX**, **ACTE**).
- Enabling **collaborative tools** that connect universities, labs, creative hubs, and networks (e.g., **European Textile Network**, **MAD Brussels**) to share knowledge and co-develop solutions.
- Supporting **transparency and traceability** to empower stakeholders such as **Ecoalf** and **Studio Sarmite** to make informed, sustainable material choices.
- Encouraging **pilot initiatives** and innovation labs to demonstrate practical, scalable circular economy models.

5. Detailed Stakeholder Insights

5.1 Universities and Schools

- **Politecnico di Milano:** Need for better industry communication and accessible data translation.
- **Aalto University:** Strong emphasis on interdisciplinary collaboration and open platforms.
- **Moholy-Nagy University:** Calls for transparent certification information and practical material databases.
- **Beuth Hochschule für Technik Berlin:** Requires verified supplier networks and sourcing support.

5.2 Creative and Textile Labs

- **Lottozero Textile Hub:** Scaling innovation beyond pilots is a key challenge.
- **Textile Prototyping Lab:** Needs better access to comprehensive supplier and certification data.
- **Waag Textile Lab:** Requests modular education and open-access knowledge repositories.

5.3 Networks and Institutional Platforms

- **European Textile Network:** Pushes for pan-European knowledge standardization and collaboration.
- **EURATEX:** Focuses on supply chain transparency and certification harmonization.
- **ACTE:** Advocates for SME access to verified material data and scalable solutions.

5.4 Creative Hubs and Studios

- **Studio Sarmite:** Limited access to affordable sustainable materials.
- **Ecoalf:** High demand for transparency and traceability tools.
- **Labore:** Emphasizes cross-disciplinary exchange.

6. Recommendations for Imasus Implementation

1. **Develop a User-Centric Digital Platform:**
Design with multi-language support and intuitive navigation to serve users ranging from **NABA students** to **Lottozero designers**.
2. **Curate and Validate Material Data:**
Partner with academic institutions such as **Aalto University** and certification bodies to ensure accuracy.
3. **Build Modular Educational Content:**
Co-create learning modules with universities and labs to cover beginner to advanced users.
4. **Enable Network and Collaboration Tools:**
Include forums and project matchmaking for entities like **European Textile Network**, **MAD Brussels**, and **ACTE** members.
5. **Support Pilot Projects:**
Facilitate funding and technical assistance for hubs like **Textile Prototyping Lab** and studios such as **Studio Sarmite**.
6. **Ensure Continuous Feedback and Adaptation:**
Engage all stakeholders in iterative development to keep the platform responsive and relevant.

7. Conclusion

The sustainable textile and fashion industry's transition hinges on **integrated knowledge sharing, transparency, and collaborative innovation**. Our dual-survey analysis, referencing concrete contributions from key institutions including **Politecnico di Milano**, **Aalto University**, **Lottozero Textile Hub**, and networks like **EURATEX**, demonstrates the complementary roles and shared challenges across the sector.

The Imasus project is uniquely positioned to bridge these gaps—by delivering an accessible, comprehensive digital ecosystem that fosters cooperation, reduces fragmentation, and accelerates the adoption of sustainable materials and circular design practices.

By responding directly to the articulated needs and ambitions of this rich stakeholder ecosystem, Imasus can become a catalyst for lasting impact across Europe's textile and fashion sectors.